MCPresso Pipeline Architecture

Target: <60s end-to-end

Success Rate: 78%

Repair Convergence: 85%

Natural Language Description

Semantic Registry Search (Similarity-based)

ADAPT ≥0.85 (~10s)

SEED 0.60-0.85 (~30s)

FULL_GEN <0.60 (~60s)

LLM Code Generator (Claude Sonnet 4.0) + Template Integration

Five-Category Validator
Structural • Protocol • Security
Robustness • Documentation

Score <75?

Repair Engine (≤3 iterations)

Composite Scorer
Validation 40% • Complexity 20%
Consistency 20% • Security 20%

Test Co-Generator
3 tests per tool

server.py + test_server.py + BrewResult